

Rough Draft to be Completed

Business Plan for the

RAAM Power & Transport

Propellantless Propulsion Transport Power Facility Development

RAM P & T Inc.



Gordon Novel
Project Director

RAMP&T.com

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Business Plan draft preparation by



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The RAM Project

Executive Summary

During the latter half of the twentieth century, the United States government back-engineered an extraterrestrial spacecraft. The information already in the public domain regarding the Alien Reproduction Vehicle (ARV), when combined with eyewitness testimony and photographic evidence, allows a strong scientific case to be made, not only for the very existence of ARV, but more importantly, for the actual circuit design and

configuration that powers it. The quantity and quality of the available information makes it feasible to back-engineer this Alien Reproduction Vehicle and, in turn, construct a working Replication ARV Machine (RAM).

The RAM Project is an elite team of prominent physicists, engineers, researchers, and entrepreneurs who are committed to designing, building, operating, and marketing air and spacecraft using propellantless propulsion systems based on zero-point energy gravitational field control and nullification. While the general public, relying upon the mainstream media for news, is largely unfamiliar with progress in this area, landmark papers published in various major scientific journals in recent years show that this cutting edge technology is entirely practical. In the white world, this is borne out, not only by the ongoing antigravity projects within Boeing, British Aerospace, Lockheed, and others, but also by working models of propellantless thrusters, such as those based on the 2001 NASA patent number 6,317,310. In the black world, this is exemplified by vehicles such as ARV.

RAM TECHNOLOGY

Modern Stochastic Electrodynamics (SED) physics theory indicates that gravity and inertia are a result of matter interacting with the zero-point radiation of the vacuum. This implies that, since gravity, inertia, and zero-point energy are all aspects of the same physical phenomenon, an electrodynamic circuit that can control gravity will also allow the energy of the zero-point to be tapped. In other words, once you have antigravity, you also have “free” zero-point energy (ZPE). The indications are that it is this link—between zero-point energy and antigravity—that is responsible, more than anything else, for the U.S. Government’s extremely tight security surrounding the entire deep black antigravity program.

The physics community recognizes the existence of zero-point energy as a proven fact. Experiments, such as those involving the Casimir effect, demonstrate the physical nature and power of zero-point energy. The problem is not to prove that zero-point energy exists (this is easily done), but rather to develop practical technologies that can tap this energy in an economical manner. Certain well-known and respected physicists who are members of the RAM Project team are at the forefront of the theoretical work being performed in this area.

The RAM Project is currently preparing to file patents that apply the latest understanding of gravity and zero-point theory to the reverse-engineered ARV drive mechanism. With appropriate funding for the proper research and development facilities, a RAM Earth transport vehicle will be constructed and ready for testing within two years, quickly followed by unmanned interstellar probes and, ultimately, manned starships.

The RAM R-XT-1, a remote controlled prototype that is the world’s first self-contained electrogravitically propelled craft able to maneuver in all three axes with no moving thrusters, has already been built and tested. A personal demonstration of the R-XT-1 will be arranged for any qualified investors who wish to see this technology in operation firsthand.

ENVIRONMENTAL IMPACT

RAM technology is essential in order not only to sustain, but also to dramatically increase the quality of human life far into the future. The availability of propellantless propulsion technology alone will be a great boon to mankind, but the truly significant product of the RAM Project is the zero-point energy technology. The term “free energy” refers to the fact that no fuel of any kind, whether chemical or nuclear in nature, is consumed during the energy production process, making zero-point energy the only truly unlimited, renewable energy source known to man.

All life on Earth will benefit from the clean energy and pollution-free transportation systems that the RAM technology portends. In addition, the drastically reduced consumption of our irreplaceable fossil fuels will ensure that humanity will have a more than adequate supply of petrochemicals for tens of thousands of years to come.

ECONOMIC IMPACT

The owners of the RAM technology have the power to shape our planet's future. One cubic foot of space contains more than enough zero-point energy to boil dry all of the oceans on Earth, and although the energy itself is “free”, the equipment to extract it is not. The enormous worldwide macroeconomic consequences of developing this technology are clearly illustrated in a recent Brazilian military white paper, *Socio-Economical Effects of the Introduction of Gravitational Control*. Within one decade, revolutionary changes in the energy and transportation industries will result in a complete reordering of the world's economic system. While no definitive world meganomics report has been prepared to date, the total world GDP is projected to increase from forty-four trillion dollars to as much as one quadrillion dollars annually. During the last century, the wealth of the world increased some fifty times; properly deployed and managed, this technology could increase the wealth of the average human being more than one hundredfold. A conservative estimate is that, during a ten-year phased technology transition period, the RAM investors can expect to earn several trillion dollars from manufacturing and utility license fees and the global equity markets.

CONCLUSION

The RAM Project will make the patented antigravity and zero-point energy technologies available to the people of the world on a licensing basis that has been carefully planned to advance human progress while preventing economic collapse. The members of the RAM Project team are deeply committed to bringing this technology out of the shadows of the black world and into the daylight of a bright new era by developing it for the benefit of all mankind.

RAM P & T (RAM) is seeking funding to build the next generation propulsion device, with knowledge gained from initial successful experiments and from solid, theoretical predictions. With initial funding of \$20,000,000., some of which is in hand, we will be able to set up a new facility and start on a series of experiments that can complete the development of an

environmentally safe, cheap and unlimited energy/propulsion source. This lab work, Phase I of the project, lasting about 20 months, will confirm the predictions that this technology can produce net energy/lift. Phase II will develop the technology to the point of a commercially viable prototype. A successful completion of Phase I will essentially assure commercial success, as no new technology will be needed for Phase II. Total funding required by RAM for Phase I and II of this project is \$220 M. RAM believes that some Phase II funding may be available from current Phase I committed sources.

RAM Propulsion can supply energy without generating radioactive materials and at far less cost than any existing energy source. Experiments performed by RAM and collaborators have already demonstrated that the technology will yield practical results. In addition, these experiments and earlier ones performed by RAM and its head physicist KH have confirmed the theory of the concept presented by RAM President, Gordon Novel. The theory predicts that commercial energy production is possible with RAM Propulsion at **costs well below 1 cent per kWh** which will compete favorably with even the best wholesale-market, off-peak, bundled electricity rates.

The new facility will allow us to optimize the efficiency of the device, to prove new theoretical predictions, and to demonstrate "break-even" (energy in = energy out) with propellantless fuel. RAM will provide a team of scientists and technicians to carry out the prototype experiments under the direction of RAM's head physicist KH. Our current and target investors will contribute substantial funds to this project as well. In addition, the [REDACTED] (with full security clearances and protocols) has agreed to help develop the prototype, provide verification and duplicate the process.

RAM is raising funds for its research program by selling non-voting shares amounting to 50% of the value of the company. Because of the exceptional nature of the technology, and the exceptional potential for economic rewards, the inventor is maintaining control over the company to prevent any possible suppression of this technology. RAM is privately offering up to 5,000,000 shares, as needed. The first 1,000,000 shares, enough to fund Phase I, will be offered at \$20. apiece. The shares will be offered in blocks of 100,000 each. Subsequent shares will be offered at a higher price. RAM estimates of return vary however we would expect that the value of capital invested in RAM will increase over time though no representation of certainties can be made at this time.

Some General Concept Background (Not RAM Science)

(The specific science used to produce the drive system will only be presented in secure environments to previously cleared science experts or panels selected by a RAM approved potential investor.)

A 1994 article in the periodical, *American Scientist*, contains a passage that is relevant to the warp drive theory.

In the May issue of *Classical and Quantum Gravity*, Miguel Alcubierre, a physicist at the University of Wales in Cardiff describes a space-travel scenario that bears an uncanny resemblance to the warp drive of science fiction. With Alcubierre's warp drive, we could reach *any* place in the universe in as short a time as we please!

The warp drive envisioned by Alcubierre is made possible by the subtleties of Einstein's general theory of relativity. According to Einstein, spacetime (the union of the three dimensions of space with the dimension of time) is not an inert substrate, but rather a dynamic entity that twists and *distorts* under the influence of *concentrations of energy*. Alcubierre suggests that it might be possible to exploit this phenomenon to travel from one star to another faster than the speed of light. This could be done by creating a disturbance in spacetime such that the region directly in front of a spaceship is contracted while the region directly behind the spaceship is expanded. This distortion of spacetime would, in effect, propel the spaceship forward like a surfer riding the crest of a breaking wave.

(*American Scientist* 1994, Vol. 82, page 422. *Emphasis mine –GN.*)

Alcubierre claims that the causal paradoxes predicted by Special Relativity in exceeding the speed of light are not relevant because light also goes along on the trip with the bird. Also, Alcubierre claims that the accelerations of millions of G-force units would not be relevant because accelerations are relative in General Relativity. An observer, if he could follow the bird departing from Earth, would view the acceleration as enormous, but our quantum leap astronauts, being weightless, would feel no instant sense of motion. Let's hope so, otherwise they would become instant goo.

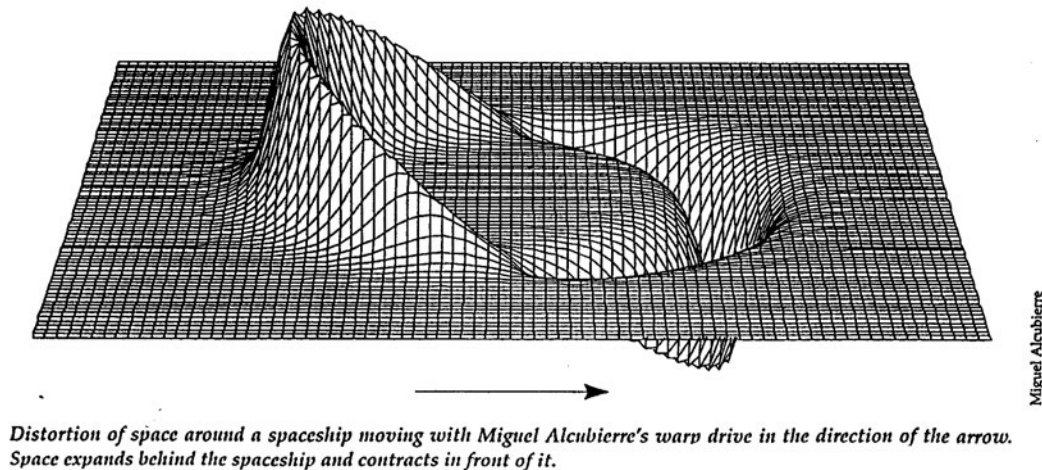


Figure 1 – Alcubierre Warp Drive Space Distortion

Another graphic in three parts illustrating how it is believed the drive system will focus its field in either the annular ring, the umbrella, or cone configuration by varying the voltage at each of the four levels of capacitors in the generator drive. The purpose of this is to increase or decrease the area of influence under the bird for the RAM generated force to push against, thus minimizing or maximizing the thrusting force.

Figure 2 – Hypothetical RAM Vehicle Ionization Patterns (Not ZPF)

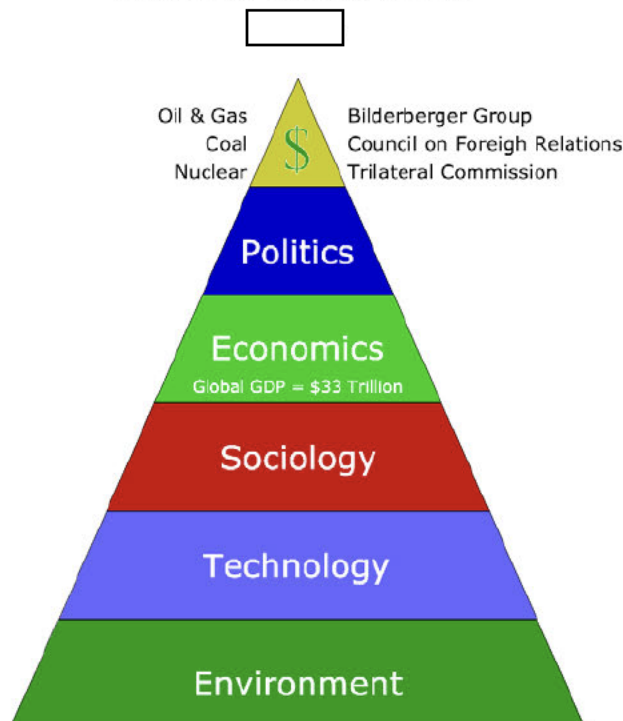
Socio Economic Consequences of RAM Technology

SOCIO-ECONOMIC SYSTEMS

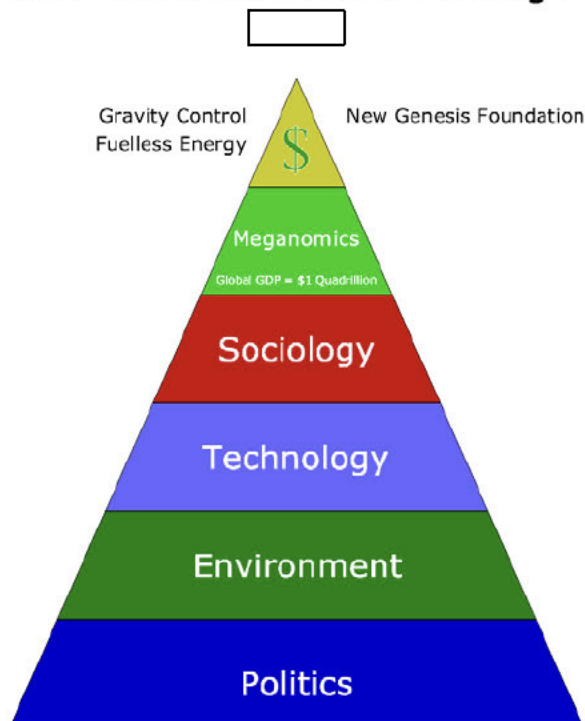
Pre-Extraterrestrial

Post Extraterrestrial

Current World Order



New World Federation Paradigm



Overview - The introduction of new advanced RAM Power & Transport Technologies:

This report will address the effects of the patented RAM P & T work product which is creating a paradigm change in physics as it relates to modern society. More precisely this new technology will address the correlation between electromagnetism and propulsion which will lead to a partial or full control a new worldwide transportation and energy model.

This report will address the technical and economical viability of an ultra efficient electric motor showing its potential as primary energy source for the next decades or centuries. This analysis will infer the expected resulting variables and the probable long term effects of the growth in energy availability and the mechanisms which will help drive toward space exploration and colonization. Finally, this RAM P & T report will probe into the risks and vulnerabilities to which our society will be exposed, and the challenges it may face as a result of this new and dynamic technology.

1. Introduction

RAM P & T has harnessed one of the basic forces of the universe, electricity, which will certainly change the world we live in. The number of technical and business opportunities that this new discovery will

open is enormous. The scope of this announcement is not to address every single possibility which will be opened with this new advance, but to focus upon the long-term effects of the introduction of an application which will have a vital impact on existing Transportation systems and which will be capable of tapping energy directly from a electrostatic fields like the one generated by Earth itself. This energy would be a new source of power and propulsion which will be magnitudes above the current models of efficiency.

2. The main hypothesis

This hypothesis is that there is the possibility of controlling the force upon a body by using electromagnetic radiation in form of waves or a light. This is an extraordinary advance in our understanding of the universe, but as any such advance it requires extraordinary evidence.

By overcoming the weight upon a body, even if partially and temporarily, it is possible to create a motion and therefore kinetic energy. This motion or motor is not perpetual since the elements in it will decay and lose performance, but it is tapping some energy source previously untapped and which has been more fully explained in the filed patent by RAM P & T.

It is important to remember that most energy sources were explored without a complete explanation for them, until late XIX century. Egyptians and Sumerians used hydraulic energy without even guessing it ultimately came from the fusion reactions within the sun by means of the radiation evaporating water and then producing rains in the mountains where the river sources were located. Many wrong theories were proposed for centuries before understanding the true nature of this energy. However, ignoring its nature didn't prevent its use.

3. Economical viability study

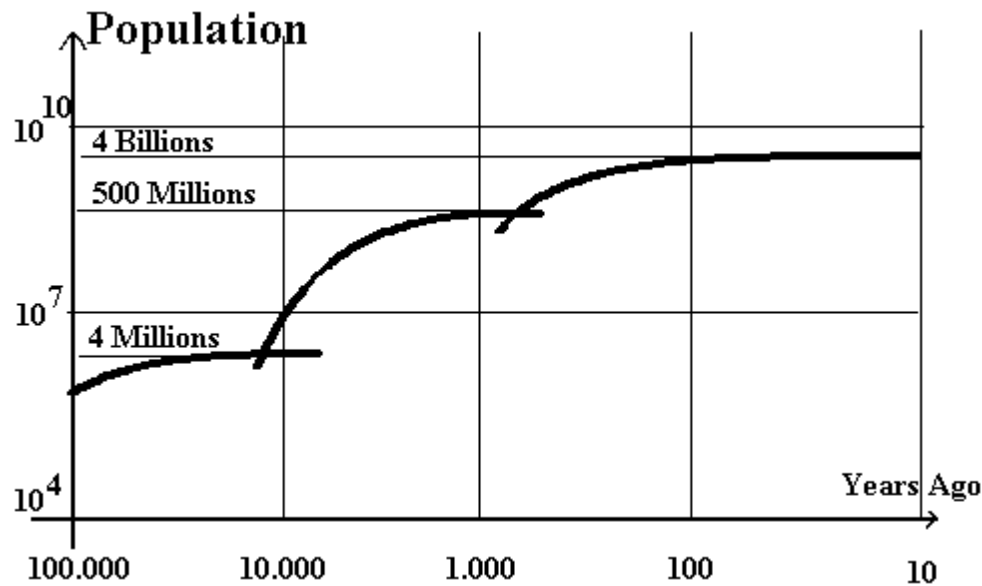
The case for the RAM P & T device/motor affords a superior system since the elements incumbent in its design decay in a much longer time, more comparable to the lifetime of a conventional electrical motor. Such motors can last decades if well maintained, but maintenance has a cost in itself. A reasonable estimate would be that it would last for around ten years without maintenance and then it would cost one tenth of its original value per year to maintain. This a rough evaluation put forth in order to enable the price of producing energy with such a motor.

4. The energy curves

Butler (1980) has shown that the growth of human population is directly correlated to energy available to human society. The Graphic I below divides human history in three parts, each directly related to a new energy source. The first curve represents the population expansion following the control of fire in the Paleolithic. The second curve is linked to the tapping of solar energy through the development of agriculture. The last part is associated with the introduction of fossil fuels in recent past.

Graphic I – Human Population along time

Source: Butler, 1980



This means that the introduction of a new and vast energy source would generate a new curve in the graphic above. In fact, the introduction of nuclear energy whether from fission or fusion would have a similar effect only to a lesser degree. We can roughly estimate a population growth by a factor of 10^2 in a few centuries. Part of this energy can be used to improve the living standard in our planet.

The most developed societies on Earth have a consumption level around 200GJ per capita per year.

The United Nations (UN/DESA, 1998) has estimated that human population on Earth will be around 11 billion people by mid 2050. This estimative doesn't suppose an energy-induced growth but rather other trend methods. To supply such population with 200GJ per capita would require 2,200 EJ (Exajoules or 10^{21} J), which means a growth by a factor of 6 from today's installed capacity of around 363 EJ. In the later part of the XX century we have doubled our capacity each thirty years which would lead us to only 1,600 EJ. This means that, not only mankind is energy dependant, but also it is starving on energy.

In these considerations we are not even taking into account the fact that our current sources of energy are either reaching their potential limit (hydroelectric) or are heavily polluting (thermal).

This could lead to our energy capacity growing much slower than the projected rate of doubling each thirty years.

Table I – World Energy consumption	Consumption, EJ
Year	
1967	172
1990	335
1997	363
2050 (projected)	1.300
2050 (needed)	2.200

5. Vulnerabilities

The human control of inertia will incur a series of vulnerabilities for the whole humankind. We cannot at this time foresee all of them, but we can look at some of the most probable or the more hazardous ones.

a. Political-military risks

We have to acknowledge that human warfare has been more damaging as more energy became available to humans. With the advent of a limitless supply of energy warfare become forbiddingly destructive. The capacity to throw an asteroid against Earth is ultimately the ability to destroy Earth civilization in one act. As these RAM propulsion & energy systems are developed their control and their use must be very strictly controlled. This is even more important than the control necessary relative to today's weapons of mass destruction.

Past experience shows that human civilizations have not refrained using "very destructive" technologies against their counterparts. Deployment of very destructive new technologies has been the case with everything from the machinegun to the atom bomb. This same experience shows that eventually the temptation for its use has proved too great a challenge to avoid. The history of the machine gun, used purely in colonial wars during XIX century, eventually became widespread in Europe during WWI and similarly the atom bomb in WWII.

b. Socio -Economic Risks

As in any other techno-economical paradigm change, some sectors of the economy will be reduced and other will grow. Other sectors will appear for the first time and others will disappear forever. This will cause great unemployment at first because whole sectors will enter a structural crisis. Only after some time, will new business and jobs appear reducing the crisis and the societal stress.

The advance of technology usually makes lots of knowledge obsolete bringing the "destruction" of intellectual capital. Many people find their knowledge and expertise suddenly obsolete and no longer desired. This is the main cause of the structural crisis following a technological breakthrough.

This change in value of intellectual capital will bring about a change in the value of many companies and will shock many nations into monopolistic economies. Many Arab nations which have economies based in oil may find themselves threatened by the breakthrough in energy development. The same will happen with States like Texas in US.

Also, the increase in energy availability may cause a disproportionate increase in supply in relation to demand generating a "post-scarcity society" as described by Giddens (1991). This will generate the effect

of many items today which are perceived as "valuable" becoming nothing more than off the shelf commodities. This has happened in the past with at least two items: salt and sugar.

Salt was very valuable in the Roman empire, and the salt mines of Sicily were regarded as one of the powerhouses of Roman economy. People were paid in salt, hence the name salarium. Sugar was a highly prized item in XVI and XVII century, it was the main product of South America and the reason why Brazil was colonized and fought over during the Thirty-years war. Both items are now reduced to mere commodities as they are now produced much beyond the needs of human population.

This "post-scarcity economy" will make the drive toward a service-based economy even stronger as services based in human relations such as tourism, art and entertainment, will always be valued relative to the human participation. The higher the living standard of a society, the more expensive and prized these services will become.

Eventually, human population will again rise bringing scarcity to some items again and rebalancing the economy. Other items may be added into our economy which will be developed as a result of being able to work more easily in space. We can forecast the introduction of new zero-gravity products built in deep space such as pharmaceuticals and other biotechnological products

6. Conclusions

The introduction of significantly increased propulsion systems and energy systems will open a new chapter in human history. It will open new possibilities and opportunities and expose humankind to new challenges and threats.

In the beginning we will find ourselves delighted with our new capacities, but soon the vulnerabilities presented before us will begin to appear in form of new warfare technologies, economic crisis, social problems and ultimately the challenge of facing a new frontier. How we manage and predict these changes will determine their positive or negative effects on our future.

Bibliography

- Socio-economical effects of the introduction of gravitational control; Paulo Vicente 2001
- Butler, Joseph H.; Economic Geography; New York, John Wiley & Sons, 1980
- De Aquino, F.; Gravitation and Electromagnetism: correlation and grand Unification; Journal of New energy, no2 pp 76-82, Los Alamos National Laboratory preprint no. qr-qc-9910036, 1999.
- De Aquino, F.; Possibility of control of the gravitational mass by means of Extra-Low Frequencies radiation; Los alamos National laboratory preprint no. qr-qc-0005107, 2000.
- De Aquino, F.; How to extract energy directly from a gravitational field; Los alamos National laboratory preprint no. qr-qc-0007069, 2000.
- De Aquino, F.; A possibility of gravity control in luminescent materials; White paper, 2001.
- Dosi, Giovanni et alli; Technical Change and Economic Theory; London, Pinter publisher's limited, 1988
- Fischer, David Hackett; The great wave; New York, Oxford University Press, 1996
- Giddens, Anthony; As consequências da modernidade, São Paulo, Editora UNESP, 1991
- Hender, T.C. et alli; Key issues for the economic viability of magnetic fusion power; UKAEA circa 1995
- Jones, Harry e Twiss, Brian; Forecasting Technology for planning decisions; New York, Macmillan Press Ltd, 1978
- Kaufmann, Stuart; At home in the universe; New York NY; Oxford University Press, 1995
- Keegan, John; A history of warfare; New york, Alfred A. Knopf, 1994
- Kennedy, Paul; The rise and fall of the great powers; Glasgow, Fontana press, 1989
- Kuhn, Thomas S.; A estrutura das Revoluções científicas; São Paulo, Editora Perspectiva, 1997
- Martino, Joseph P.; Technological Forecasting for Decision Making; New York, Elsevier, 1972
- Mokyr, Joel; Twenty-Five Centuries of Technological change: an Historical Survey; London, Harwood Academic publishers, 1990

Puthoff, H. E.; Gravity as a Zero-point fluctuation force; Physical review A, volume 39, number 5 page 2333, 1989.

Ribeiro, Darcy; O processo civilizatório; São Paulo, Circulo do livro , 1978.

Roberts, J. M.; History of the world; New York, Oxford University Press, 1993

Schumpeter, Joseph A.; The Theory of Economic Development; New York, Oxford University press, 1961

Wesson, Paul S.; Zero-Point fields, gravitation and new physics; White paper, 2000

Internet Resources

Hatcher, Stanley R.; 2050: Nuclear power in the next century; www.chem-eng.utoronto.ca/~che3000/UOT9812.html, 1998

Hardeman, C.; The Aquino-Hardeman photo-gravity effect; www.icnet.net/users/chrish/Photo-gravity.htm, 2001.

Millis, M.; NASA Breakthrough propulsion physics program; www.grc.nasa.gov/WWW/bpp/TM-1998-208400.htm, 1998

Puthoff, H. E.; Can the Vacuum be Engineered for Spaceflight applications?; www.keelynet.com/gravity/putnasa.htm, 1997

Schneider, Jean; The extrasolar planets Encyclopaedia; <http://www.obspm.fr/encycl/encycl.html>, 2001
UN/DESA, Population Division; World population projections to 2150; www.undp.org/popin/wdtrends/execsum.htm, 1998

Uranium Information Center; The economics of Nuclear power; www.uic.com.au/nip08.htm, january 2001

Watmough, John; Extrasolar visions; <http://www.jtwinc.com/Extrasolar/>, 2000

Advantages of RAAM Propulsion

RAM Drive Devices will have several great advantages over existing energy sources:

1. **RAAM Propulsion devices are safe and environmentally sound.** No long-term radioactive by-products or pollutants are produced. The end-product is harmless. RAAM Propulsion reactors would be free of radioactivity and the small number of low-energy neutrons emitted can be easily absorbed in several inches of shielding.
2. **RAAM Propulsion reactors are cheap.** Almost all of the energy is released in the motion of charged particles. RAAM Propulsion-generated electricity costs are projected to be as much as one hundred times less than present energy costs. Mass production would reduce prices substantially. In comparison, the realized future tokamak design of fusion reactor, due to the requirement for a steam generator, cannot reduce electricity costs at all.
3. **RAAM Propulsion reactors are small and decentralized.** RAAM Propulsion reactors can fit into a residential garage and can be made as small as 2 MW, sufficient for a small community.

Comparison of RAAM Propulsion to the Tokamak

Reactor Type	RAAM Propulsion	Tokamak
Fuel	Various Proprietary	Deuterium-tritium
Fuel availability	Abundantly available	Tritium must be bred

Long-lived radioactivity	None	Considerable
Radioactivity of structure	None	Considerable
Power output per unit	2 MW and up	500 MW and up
Unit size	3x3x3 feet	70x70x80 feet
Capital Cost per kW	\$10 - \$200	\$2000 – 3000
Electricity conversion	Direct induction	Steam cycle

How the Plasma Focus Works

Operation

RAM Density Measurement

Energy Conversion in RAAM Propulsion

Energy Flow Sequence Diagram

Feasibility of RAAM Propulsion

Project Objectives

Project Tasks

Task 1. Purchase of equipment.

This task involves the purchase of

Task 2. Theoretical calculations and design of

Task 3. Assembly of Facility.

Task 4. Planning and Preparation for future experimental development stage

Simultaneously with Task 3, while the facility is being assembled, RAM will be planning and making preparations for the Phase II development experiments, so that there will be no break in work following completion of the first experimental tests. These plans will of course be refined on the basis of the initial experimental results.

Task 5. Testing of facility and calibration of instruments

Once the facility is fully assembled, RAM Physicists and Technicians will carry out a series of preliminary tests to shake down the facility and to calibrate all the instruments.

Task 6. First experimental set

The first set of confirmations will demonstrate the theory that higher efficiency of energy transfer into the hot spot, can be achieved with

Task 7. Second experimental set

Task 8. Third experimental set

Task 9. Fourth experimental set

Task 10. Preparation of papers for publication.

Phase I Project Schedule – Gantt Chart

Task	Duration	1	2	3	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	6 Months																			
2	6 Months																			
3	3 Months																			
4	3 Months																			
5	2 Months																			
6	1 Month																			
7	1 Month																			
8	2 Months																			
9	3 Months																			
10	2 Months																			
Total	20 Months																			

Phase I Project Budget

Task	Equipment	Labor	Travel	Overhead	Total
1	\$	\$	\$	\$	\$
2					
3					
4					
5					
6					
7					
8					
9					
10					
Total					

Note: While this budget is drawn up for the 20 month task schedule, the Combined Budget in the Appendix covers two full years for Phase I.

Phase II

In Phase II, extending through years 2.5.3, 4 and 5, RAM will develop the technology demonstrated in Phase I into a working prototype fusion generator with approximately _____ output. This will involve optimizing _____, switching and conditioning technology, testing the technology at the high repetition rates needed for efficient functioning. The budget for this period is approximately \$18 million. However, RAM anticipates that there is a good chance that not all of this money needs to be raised from additional investors. There is a good possibility that once break-even is achieved, funding from private sources either in the US or in Europe will be forthcoming for the rest. Such contracts would enable RAM to operate at a profit even before development is completed.

Phase III

In Phase III, RAM would market the RAAM Propulsion devices. We believe that the fastest and lowest-risk method of doing this is through selling non-exclusive licenses on the technology. RAM will be protecting its intellectual property rights with a series of patents. Likely initial licenses will be governmental agencies of oil-importing countries, such as, for example Japan, France and Italy. The sale of such licenses will generate a relatively large income stream initially that will be supplemented when royalties begin to flow after actual production is begun. In the longer term, RAM may, with some of the accumulated revenue, proceed to establish its own manufacturing facilities either independently or in joint ventures.

Market for RAAM Propulsion Energy

Distributed Generators

RAM engineering analysis indicates that 2 MW RAAM Propulsion reactors could be produced for about \$200,000 apiece. This is about one-tenth of the cost of conventional electricity generation units of any style or fuel design. This means that once the prototype is successfully developed within five years, RAAM Propulsion generators will be the preferred technology for new electrical distributed generation

More powerful units can be designed by accelerating the pulse repetition rate, although there are limitations due to the amount of waste heat that can be removed from such a small device. It is likely that units larger than 20MW will be formed by simply stacking smaller units together, with approximately the same cost per kW of generated power or by scaling up in size.

We can project the eventual market for new electric generators. Current global new electric generation capacity today amounts to about 100 GW per year, averaging over the last decade. We estimate that the introduction of a much cheaper energy source will in fact increase growth of electric consumption considerably. There will also be a significant market for the replacement of existing sources as well.

The conservative estimate of eventual market size can be used to estimate income stream for RAM. Assuming a price of \$100 per new kilowatt of installed RAAM Propulsion power generators, a royalty of 5% we have an eventual revenue stream of \$500 M per year. By the same reasoning, a 10% market penetration of the new electricity generation market yields an income stream of \$1,000 M per year.

An additional important source of income is from the initial sale of the licenses themselves, even before royalties are forthcoming. We expect that, given the size of the market and the importance of the technology, initial payments on five or six non-exclusive licenses will be in excess of \$10 million apiece. This is very conservative, since a license that leads to 10% market penetration will generate \$300 million/ year in profit, assuming a modest 30% net profit on sales.

Space Propulsion

Another market that is available to this product is the space propulsion market. Nuclear propulsion is a hot subject, recently reviewed in [New Scientist magazine](#) (Jan. 20 & 23, 2003) with NASA's Nuclear Systems Initiative being renamed "Project Prometheus" and an increased budget recently approved by the White House. NASA explains that 600 million degrees was a prerequisite for this modality but that 6-8 weeks may be possible for a trip to Mars with a tripling of the space travel speed. The development of RAM Technology for space propulsion has been, for many years, a long term goal of the space program. However, the difficulty of achieving this power generally due to the very low thrust-to-weight ratios of most propellantless propulsion designs has made this goal appear impracticable.

But in the past few years, there has been a growth of interest in the propellantless fuel devices as a possible space propulsion system.

Potentially RAAM Propulsion thrusters will have very high specific impulse, compared to chemical rockets which have very low exhaust velocities of only 2 km/sec that are obtainable in this manner. To achieve minimal space velocities above 25 km/sec need for interplanetary travel,

far more thrust than payload is required. In contrast, the beam from a RAAM Propulsion device exits at over 7,000 km/sec.

Over time, RAM propulsion devices could make possible robotic interstellar probes. While fusion rockets would be limited to about one third the speed of light, even if they were large with many stages, a space effort willing to sustain projects of over several decades (the cathedrals took far longer) could undertake 60-70 year missions to nearby earth-like planets, once they had been identified by astronomers. Such identification could happen in the next 10-15 years using instruments already being developed by NASA.

This new propulsion generation market may yield an additional income stream of \$25 M per year, depending on the thruster price that the market may bear. This excludes income that may be generated from the initial sale of the licenses themselves and the distributed generator income cited above.

Income Projections and ROI

Based on the above market considerations, RAM has developed an income projection assuming that substantial private funding will be available for Phase II. In this case we assume all shares are sold at \$100 initially, although we expect that after Phase I is completed, that newly issued shares will be sold for a substantially higher price. The return on investment (ROI) is calculated as an average annual rate for an investor buying shares at the start of year 1, paying \$100 a share, a price which is guaranteed only for the first 1,000,000 shares issued, and holding until the end of the cited year 2. We assume a P/E ratio of 30 in calculating the value of the company, which is conservative for a rapidly growing high-tech company that is actually turning a profit.

Projection

Year	3	4	5	6	7	8	9	10
Income (\$ millions)								
fees	0.3	0.3	0.3					
License Sales				10	10	10	10	20
Royalties					2.5	7.5	18	40
Total	0.3	0.3	0.3	10	12.5	17.5	28	60
Company Value	9	9	9	300	375	525	840	1800
Value per share (\$ thousands)	0.28	0.28	0.28	9.3	12.7	16.4	26.3	56.2
%ROI per annum	41	29	23	113	100	89	86	88

As can be seen, even with quite conservative assumptions, annual average ROI will be in the area of 90% per annum.

On the basis of this large ROI, RAM feels that non-voting shares will be an excellent investment. We are selling only non-voting shares simply because we believe the risk of efforts to suppress this technology is high, and selling voting shares will make it too easy for others to take over and suppress RAAM Propulsion. As a result, all voting shares will rest with the core investors and inventors.

Risk factors

*All new technology development programs involve risk, which is compensated for by expected high rates of return in the event of success. In our case, RAM locates the largest risk in Phase I of the project. While the theoretical projection RAM has made strongly indicates that new energy production can be achieved with RAM Propulsion in an extremely economical manner, and these theoretical models have been tested by experiment and are based on firmly established physical principles, considerable extrapolation is involved. It is always possible that unforeseen factors may prevent the achievement of break even, although RAM considers this unlikely. There are no **known** physical problems that would prevent achieving net energy production.*

Once break even is achieved the risks involved in Phase II are much lower, and in fact are, in our view, negligible. Mature technologies are available that can be tailored to the task of efficiently extracting and converting energy emitted by the RAAM Propulsion device.

In Phase III, there are some risks that fossil fuel interests, through their influence on the governments of oil-producing nations will attempt to create unjustified regulatory barriers in the way of marketing RAM Propulsion reactors. However, we believe that the extremely safe and environmentally benign nature of RAM Propulsion, together with its great economy, will produce sufficient political pressure to overcome such barriers. In any case, we believe that there will be many non-producing, oil-importing nations who will be eager to purchase and license this technology. So overall, RAM believes that, once breakeven is achieved at the end of Phase I, the risk that the technology will not achieve a high market penetration are very low.

Competition

The most successful magnetic confinement tokamak is the JET (Joint European Torus) in Culham, England. This \$1 billion machine is most powerful tokamak today and the last in a long line of attempts in many countries over the last 40 years to achieve fusion. The JET is still not expected to produce more fusion energy than it consumes and the US DOE projects another 35 years before a commercial prototype may be ready.

The most prominent inertial confinement laser fusion experiment is the NIF (National Ignition Facilities) at the Lawrence Livermore National Labs. This \$3 billion machine uses high powered lasers to heat and compress a small sphere of fusible material to high temperatures and pressures. It is not expected to produce economical energy even if it ever reaches breakeven.

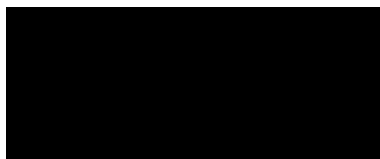
The most important electricity provider today is the fossil fuel power plants, capturing 90% of the market, which are the main competitor for RAAM Propulsion. However, when the 2 MW RAAM Propulsion reactors come on the market, they will completely displace the oil, coal, and natural gas power plants within a few years because of the substantially lower capital cost, ease of use, lack of appreciable fuel cost, and the complete lack of harmful pollution to the air, water, or ground. Therefore, even fossil fuel power generation plants do not present any real competition to RAAM Propulsion.

Project Personnel

Gordon Novel

CEO

California



EDUCATION

WORK EXPERIENCE

RESEARCH

To be named

BOOK

ORIGINAL SCIENTIFIC PAPERS

PATENTS

AWARDS

PROFESSIONAL SOCIETIES

APPENDIX

RAM P & T 5-Year Budget (Must be Reworked)

RAM Combined Budget For Phases I, II and III											
\$ x 1000	Year1	Year2	Year3	Year4	Year5	Year6	Year7	Year8	Year9	Year10	Total
Revenue											
Gov't fees			300	300	300						900
-						10000	10000	10000	10000	20000	60000
Royalties							2500	7500	18000	40000	68000
Total	0	0	300	300	300	10000	12500	17500	28000	60000	128900
Costs											
Payroll											
Salaries	1000	1000	1200	1200	1200	1800	1800	2000	2000	2000	13440
Fringe	500	500	240	240	240	360	360	400	400	400	2688
Total	1500	1500	1440	1440	1440	2160	2160	2400	2400	2400	16128
Period											
Travel	2	3	10	25	25	25	25	25	25	25	190
Leasehold Improvements	0	0	35	0	0	0	0	0	0	0	35
Facility Rent	0	1	60	60	60	60	60	60	60	60	481
Utilities & Taxes	1	1	30	30	30	30	30	30	30	30	242
Documentation	1	1	5	5	5	5	5	5	5	5	42
Office Equipment	2	1	100	20	20	20	20	20	20	20	243
Shipping	1	1	2	2	2	2	2	2	2	2	18
Insurance	0	0	1	1	1	1	1	1	1	1	8
Marketing	0	0	0	0	10	10	10	10	10	10	60
Maintenance / Supplies	1	1	8	10	15	15	15	15	15	15	110
Lab Equipment	75	10	200	150	100	50	50	50	50	50	785
Electrical Supplies	1	1	150	100	100	50	50	50	50	50	602
Machine Shop Equipment	1	1	150	50	50	0	0	0	0	0	252

Parts & Fabrication	5	5	90	100	100	50	50	50	50	50	550
Contingency	1	2	20	20	20	20	20	20	20	20	163
Total Period Cost	91	28	861	573	538	338	338	338	338	338	3781
Total Cost			2301	2013	1978	2498	2498	2738	2738	2738	19909
Profit/ Loss			-2001	-1713	-1678	7502	10002	14762	25262	57262	108991
Cash Flow			-2408	-4121	-5799	1703	11705	26467	51729	108991	

History of the RAAM Propulsion Development

Articles of Support